Lines and Shapes

Lines, rays, parallel and perpendicular, and lines of symmetry.

Segment, ray, line



line segment: 2 endpoints



ray: 1 endpoint

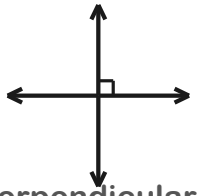


line: no endpoints

Parallel and perpendicular

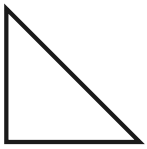


Parallel: never meet.



Perpendicular: a square corner.

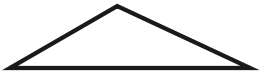
Triangles by their biggest angle



right

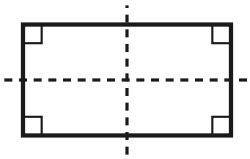


acute



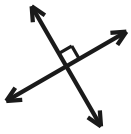
obtuse

Lines of symmetry: fold it and the halves match



A rectangle has 2 lines of symmetry.

Your turn!



parallel perpendicular neither

Lines and Shapes: Practice

Look for endpoints, square corners and matching halves.

Part 1 Segment, ray or line? Count the endpoints.

1)  segment ray line

2) segment ray line

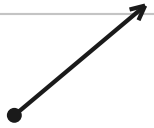
3)  segment ray line

4)  segment ray line

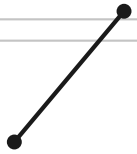
5)  segment ray line

6)  segment ray line

7)  segment ray line

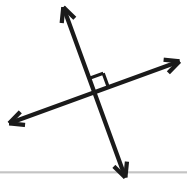
8)  segment ray line

9)  segment ray line

10)  segment ray line

Lines and Shapes: Practice (continued)

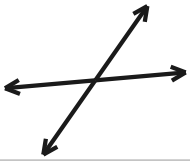
Part 2 Parallel, perpendicular or neither? Circle one.

11) 

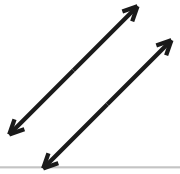
parallel perpendicular neither

12) 

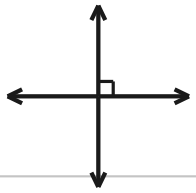
parallel perpendicular neither

13) 

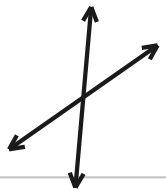
parallel perpendicular neither

14) 

parallel perpendicular neither

15) 

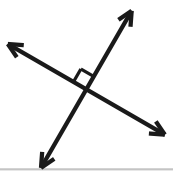
parallel perpendicular neither

16) 

parallel perpendicular neither

17) 

parallel perpendicular neither

18) 

parallel perpendicular neither

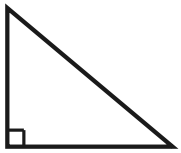
Lines and Shapes: Practice (continued)

Part 2 Parallel, perpendicular or neither? (continued)

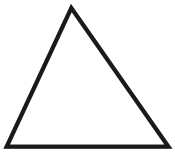
19) 

parallel perpendicular neither

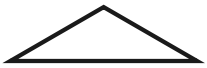
Part 3 Right, acute or obtuse triangle? Look at the biggest corner.

20) 

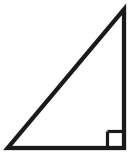
right acute obtuse

21) 

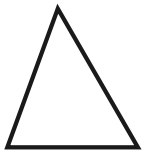
right acute obtuse

22) 

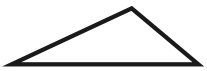
right acute obtuse

23) 

right acute obtuse

24) 

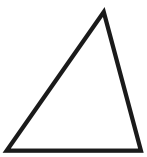
right acute obtuse

25) 

right acute obtuse

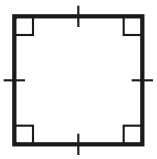
26) 

right acute obtuse

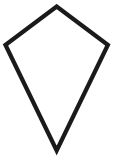
27) 

right acute obtuse

Part 4 Lines of symmetry How many ways can you fold it so the halves match?

28) 

lines of symmetry: ☐

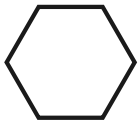
29) 

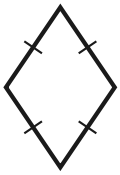
lines of symmetry: ☐

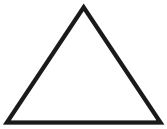
Lines and Shapes: Practice (continued)

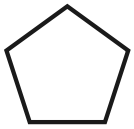
Part 4 Lines of symmetry (continued)

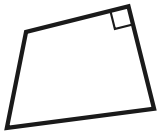
30)  lines of symmetry: ☐

31)  lines of symmetry: ☐

32)  lines of symmetry: ☐

33)  lines of symmetry: ☐

34)  lines of symmetry: ☐

35)  lines of symmetry: ☐

Part 5 True or not? Circle yes or no.

36) **A square has 4 right angles.** yes no

37) **Parallel lines cross.** yes no

38) **A line has two endpoints.** yes no

39) **A ray has one endpoint.** yes no

40) **A rectangle has 2 pairs of parallel sides.** yes no

Lines and Shapes: Practice (continued)

Part 5 True or not? (continued)

41) **A right triangle has 2 right angles.** yes no

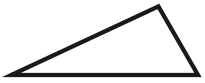
42) **Perpendicular lines make square corners.** yes no

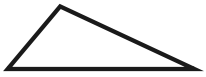
43) **A circle has no lines of symmetry.** yes no

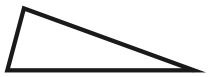
Part 6 What kind of triangle?

44)  right acute obtuse

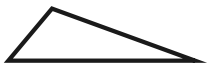
45)  right acute obtuse

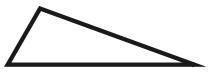
46)  right acute obtuse

47)  right acute obtuse

48)  right acute obtuse

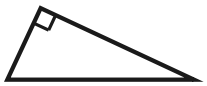
49)  right acute obtuse

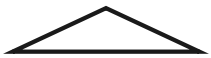
50)  right acute obtuse

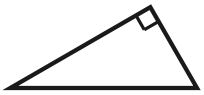
51)  right acute obtuse

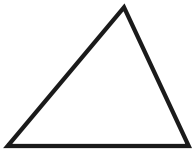
Lines and Shapes: Practice (continued)

Part 6 What kind of triangle? (continued)

52)  right acute obtuse

53)  right acute obtuse

54)  right acute obtuse

55)  right acute obtuse

Part 7 Remember? Problems from earlier packets.

56) $40 + \square = 90$ degrees

57) $2\frac{1}{8} + \frac{5}{8} = \square\frac{\square}{8}$

58) area: $21 \times 4 = \square$ sq m

59) 7 ft $= \square$ in

60) $0.19 = \frac{\square}{100}$

61) $63 \times 22 = \square$

Lines and Shapes: Practice (continued)

Part 8 Word problems Draw a picture.

62) A window has 4 square corners. How many right angles does it have?

Number sentence _____

Answer ☐ right angles

Draw or show your work

63) Two train tracks never meet. Are they parallel or perpendicular?

Number sentence _____

Answer ☐

Draw or show your work

64) A stop sign has 8 equal sides. How many lines of symmetry does it have?

Number sentence _____

Answer ☐ lines

Draw or show your work

65) Challenge: A triangle has corners of 35 and 45 degrees. Is it acute, right or obtuse?

Number sentence _____

Answer ☐

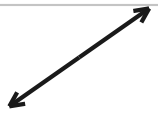
Draw or show your work

Check-In: Lines and Shapes

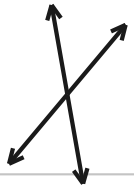
Show what you know. No help this time. Score: _____ / 8

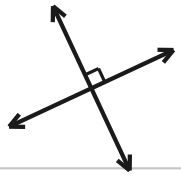
Part 1 Segment, ray or line?

1)  segment ray line

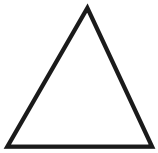
2)  segment ray line

Part 2 Parallel, perpendicular or neither?

3)  parallel perpendicular neither

4)  parallel perpendicular neither

Part 3 Right, acute or obtuse triangle?

5)  right acute obtuse

6)  right acute obtuse

Part 4 True or not? Circle yes or no.

7) **An obtuse angle is more than 90 degrees.** yes no

8) **A rhombus always has a right angle.** yes no